COBALT RIDGE AUTOMATION · COMPONENT ANALYSIS LABORATORY

Milwaukee, WI · Report CAL-2026-0731 · Issued 21 April 2026

Returned Component Inspection Report

Preliminary · Analysis in progress · Not for external distribution

1. Sample identification

Field	Value
Component	Drive assembly, CR-DRV-4410-B (Revision B)
Serial	4410B-22781
Removed from	Sortation Line SR-440, Northstar Retail Distribution, Joliet DC
Removed by	Meridian Industrial Services (third party), 6 April 2026
Received at laboratory	16 April 2026, RMA CRA-RMA-2026-0442
Related case	WR-2026-0417
Sample position in study	Unit 4 of 4
Disposition	HOLD

2. Purpose

This unit is the fourth returned drive assembly of the same revision examined under the open review of thermal failures in the SR-440 drive family. The purpose of this inspection is to determine whether a common failure mechanism is present across the returned population, and whether that mechanism is attributable to component design, manufacture, supplier variation, installation, or operating conditions.

3. Visual examination

Heat discoloration is present on the heatsink face adjacent to the power stage, extending approximately 40mm from the centerline in a broadly symmetric pattern. Discoloration ranges from straw to light blue, consistent with sustained elevated temperature rather than a single thermal excursion.

No mechanical damage, impact marks, or evidence of liquid ingress. Mounting face is flat and shows normal witness marks. Connector housings are intact and correctly seated. Conformal coating is intact except in the discolored region, where it shows slight yellowing.

4. Measurements

Measurement	This unit	Units 1-3 (range)	New reference
Thermal interface thickness, mean (mm)	0.31	0.28 - 0.34	0.25 nominal
Heatsink flatness deviation (mm)	0.04	0.03 - 0.06	< 0.08 spec
Power stage junction resistance (mOhm)	4.7	4.4 - 5.1	3.9 typical

Fan assembly free-run (rpm)	3,180	3,110 - 3,240	3,200 nominal

Thermal interface thickness on all four returned units sits above the nominal value. The deviation is within the supplier's stated tolerance band but clusters at the upper end across every unit examined.

Figure 1 · Power stage junction resistance by unit (mOhm)

Rejection limit 6.0 mOhm (not shown). Unit 4 is the sample covered by this report. No returned unit exceeds the rejection limit individually.

5. Observations

1. All four returned units show the same discoloration pattern in the same location.
2. All four show elevated junction resistance relative to a new reference unit, though none exceed the rejection limit taken individually.
3. Three of the four units, including this one, came from sites where the recorded ambient in the equipment aisle exceeded 27C during the reported failure period.
4. This unit was operated for an undetermined period with the thermal cutback threshold raised above the released baseline value, per the third-party work order associated with the site. The duration at the elevated threshold is not established.

> **Analyst comment.** The pattern across four units is consistent with a marginal thermal design or a supplier process variation at the thermal interface, but the sample is small and three of the four sites share an elevated ambient condition. The evidence does not yet separate a component-attributable cause from an installation-and-environment cause. A conclusion should not be drawn from this report alone.

6. Outstanding work

- Cross-section of the power stage on units 2 and 4 to examine solder joint condition. Scheduled 28 April.
- Supplier lot trace for all four units. Requested from Procurement 17 April, not yet received.
- Ambient temperature data for the fourth site over the ninety days preceding failure. Requested from the customer, not yet received.
- Comparison against a control population of units from the same lots still in service.

7. Interim conclusion

Preliminary. No common failure mechanism is established at this time. The unit is retained on hold pending completion of Section 6. This report does not support a coverage determination on case WR-2026-0417 in either direction, and should not be cited as establishing a product defect.

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Reviewed · P. Shah, Product Quality Lead